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Electric vehicle charging connector adapter nested in an electric vehicle supply equipment

Abstract

A charging connector adapter is nested within a holster of an electric vehicle supply equipment (EVSE). If the adapter is to be used, the EVSE automatically locks the charging connector to the charging connector adapter and unlocks the charging connector adapter from the holster of the EVSE thereby allowing the charging cable with the charging connector adapter to be removed from the holster and used for connecting the charging connector adapter to an inlet of an electric vehicle. When the charging connector adapter is inserted back into the holster, the EVSE locks the charging connector to the holster and unlocks the charging connector from the charging connector adapter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/585,392, filed Sep. 26, 2023, which is hereby incorporated by reference.

FIELD

(1) Embodiments of the invention relate to the field of electric vehicle charging; and more specifically, to an electric vehicle charging connector adapter nested in an electric vehicle supply equipment.

BACKGROUND

(2) There are different electric vehicle charging connector types that take different physical forms. Example connector types include: IEC 62196 Type 1 (defined in IEC 62196-1:2022, published May 3, 2022, and sometimes referred to as SAE J1772), IEC 62196 Type 2 (defined in IEC 62196-2:2022, published Oct. 19, 2022, and sometimes referred to as Mennekes), Combined Charging System (CCS) Combo 1 (CCS1) (defined in IEC 62196-3:2022, configuration EE), CCS Combo 2 (CCS2) (defined in IEC 62196-3:2022, configuration FF), CHAdeMO (defined in IEC 62196-3:2022, configuration AA), and North American Charging Standard (NACS). An electric vehicle typically has an inlet that is designed to use a particular charging connector type. Different charging protocols may exist for different connector types. Standalone adapters exist to adapt from one connector type to another.

SUMMARY

(3) In one aspect, a method in the EVSE is performed including locking a charging connector adapter to a holster of the EVSE, where the charging connector adapter is nested within the holster of the EVSE; determining that an electric vehicle that is to be charged at the EVSE through a charging cable with a charging connector is to use the charging connector adapter; automatically locking the charging connector to the charging connector adapter; automatically unlocking the charging connector adapter from the holster of the EVSE thereby allowing the charging cable with the charging connector adapter to be removed from the holster and used for connecting the charging connector adapter to an inlet of the electric vehicle; and responsive to the charging connector adapter being inserted back into the holster, locking the charging connector adapter to the holster of the EVSE, and unlocking the charging connector from the charging connector adapter.

(4) The charging connector may be Combined Charging System (CCS) Combo 1 compliant, and the charging connector adapter may be adapted to connect to an electric vehicle inlet that is North American Charging Standard (NACS) compliant. Automatically locking the charging connector to the charging connector adapter may include deactivating an electromagnet that moves a locking pin of the charging connector adapter to a locked position thereby preventing a latch of the charging connector to be unlatched, where the electromagnet is external to the charging connector adapter. Automatically locking the charging connector to the charging connector adapter may include using a linear actuator to disengage a locking pin of the charging connector adapter with a ball-lock quick release pin to move the locking pin to a locked position thereby preventing a latch of the charging connector to be unlatched. Automatically locking the charging connector to the charging connector adapter may include using a linear actuator to move a first magnet of the holster to a position where a magnetic force of the first magnet does not attract a second magnet that is attached to a latch of the charging connector adapter thereby preventing a latch of the charging connector to be unlatched. Automatically unlocking the charging connector adapter from the holster of the EVSE

may include retracting a spring-loaded latch that latches to the charging connector adapter. Determining that the first electric vehicle that is to be charged at the EVSE through the charging cable with the charging connector is to use the charging connector adapter may include receiving a communication that specifies or indicates an intention to use the charging connector adapter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention may best be understood by referring to the following description and accompanying drawings that are used to illustrate embodiments of the invention. In the drawings:
- (2) FIG. 1 shows a portion of an electric vehicle supply equipment (EVSE) that includes an adapter that is nested within the EVSE according to an embodiment.
- (3) FIG. 2 is a front view of an exemplary charging connector according to an embodiment.
- (4) FIG. 3 is a perspective side view of the exemplary charging connector of FIG. 2.
- (5) FIG. 4 is a perspective view of a first side of the adapter according to an embodiment.
- (6) FIG. 5 is a front view of the first side of the adapter.
- (7) FIG. 6 is a perspective view of a second side of the adapter according to an embodiment.
- (8) FIG. 7 is a front view of the second side of the adapter.
- (9) FIG. 8 is a first view of the internal components of the adapter for electrically connecting the conductive components of the first side of the adapter with the contacts of the second side of the adapter according to an embodiment.
- (10) FIG. 9 is a second view of the internal components of the adapter shown in FIG. 8.
- (11) FIG. 10 is a perspective view of the adapter connected to the connector according to an embodiment.
- (12) FIG. 11 is a side view of the adapter connected to the connector shown in FIG. 10.
- (13) FIG. 12 shows a holster arrangement that includes two holsters that are each capable of holding a connector when not in use according to an embodiment.
- (14) FIG. 13 shows a side view of a holster according to an embodiment.
- (15) FIG. 14 shows the side view of the holster of FIG. 13 without the holster housing.
- (16) FIG. 15 shows a cross-section view of the side view of FIG. 13 when the latch is in a locked position.
- (17) FIG. 16 shows a cross-section view of the side view of FIG. 13 when the latch is in an unlocked position.
- (18) FIG. 17 illustrates a portion of a cross-section of the connector, adapter, and holster that shows the locking control using a locking pin contained within the adapter controlled by the electromagnet that is external to the adapter, according to an embodiment.
- (19) FIG. 18 shows the locking pin plunger shown in FIG. 17 pulled back by the electromagnet thereby putting the locking pin in an unlocked position.
- (20) FIG. 19 shows the locking pin plunger shown in FIG. 17 in a locking position thereby putting the locking pin in a locked position.
- (21) FIG. 20 shows a cross-section view of the connector, the adapter, and the holster according to an embodiment.
- (22) FIG. 21 shows the cross-section view of FIG. 20 where the ball-lock quick release pin engages the adapter locking pin and overcomes the spring force of the compression spring to move the adapter locking pin to an unlocked position.
- (23) FIG. 22 shows the cross-section view of FIG. 20 where once a hard stop is hit, the ball-lock quick release disengages from the adapter locking pin.
- (24) FIG. 23 shows the cross-section view of FIG. 20 where after disengaging, the compression spring returns the adapter locking pin to a locked position.

- (25) FIG. 24 shows a perspective view of the first side of the adapter according to an embodiment where the adapter is locked and unlocked from the connector using a latch with an attached magnet.
- (26) FIG. 25 shows an exemplary latch according to an embodiment of FIG. 24.
- (27) FIG. 26 shows a cross-section view of the adapter of FIG. 24 according to an embodiment.
- (28) FIG. 27 shows a cross-section view of the connector, adapter, and the holster that shows the locking control using a latch with an attached magnet according to an embodiment.
- (29) FIG. 28 shows the latch in an unlocked position thereby allowing the connector to be removed from the adapter according to an embodiment.
- (30) FIG. 29 shows the latch in a locked position (the magnetic force from the magnet is not present) thereby locking the connector to the adapter according to an embodiment.
- (31) FIG. 30A shows an exemplary circuit schematic according to an embodiment.
- (32) FIG. 30B shows a circuit that replaces a Zener diode of the exemplary circuit of FIG. 30A according to an embodiment.
- (33) FIG. 31 is a table that shows exemplary states according to an embodiment.
- (34) FIG. 32 is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment.
- (35) FIG. 33 is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment.
- (36) FIG. 34 is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment.
- (37) FIG. 35 is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment.
- (38) FIG. 36 is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment.
- (39) FIG. 37 shows an exemplary embodiment of an EVSE according to an embodiment.

DESCRIPTION OF EMBODIMENTS

(40) An electric vehicle charging connector adapter (“adapter”) is nested within an electric vehicle supply equipment (EVSE). The EVSE has an attached charging cable where one end is secured at the EVSE and the other end terminates with a first connector type that is designed for use for electric vehicles with a corresponding inlet type. The EVSE includes a component that holds the charging cable and connector when not in use. This component is sometimes referred herein as a “holster.” The adapter is for a second connector type that is designed for electric vehicle inlets that correspond with the second connector type. The adapter is nested and locked within the holster. When the adapter is to be used (e.g., the electric vehicle to be charged has an inlet that corresponds with the second connector type), the adapter is locked to the connector of the charging cable and the adapter is unlocked from the holster. When the adapter is not to be used (e.g., the electric vehicle to be charged has an inlet that corresponds with the first connector type), the adapter is not locked to the connector.

(41) In an embodiment, the adapter is locked to the connector when the connector is connected to the holster in an idle state. In such an embodiment, if the adapter is not to be used (e.g., the electric vehicle to be charged has an inlet that corresponds with the first connector type), the adapter is automatically unlocked from the connector. The adapter may remain locked to the holster. The vehicle operator may then remove the connector from the holster and plug the connector into their electric vehicle. Responsive to the connector being inserted back into the holster, the adapter is automatically locked to the connector. If, however, the adapter is to be used (e.g., the electric vehicle to be charged has an inlet that corresponds with the second connector type), the adapter is unlocked from the holster and the vehicle operator may then remove the connector with the attached adapter from the holster and plug the adapter into their electric vehicle. Responsive to the connector with the attached adapter being inserted back into the holster, the adapter is locked to the

holster.

(42) In another embodiment, the adapter is not locked to the connector when the connector is connected to the holster in an idle state. In such an embodiment, if the adapter is not to be used (e.g., the electric vehicle to be charged has an inlet that corresponds with the first connector type), the vehicle operator may remove the connector from the holster and plug the connector into their electric vehicle. The adapter may remain locked to the holster. If, however, the adapter is to be used (e.g., the electric vehicle to be charged has an inlet that corresponds with the second connector type), the adapter is automatically locked to the connector and the adapter is automatically unlocked from the holster. The vehicle operator may then remove the connector with the attached adapter from the holster and plug the adapter into their electric vehicle. Responsive to the connector with the attached adapter being inserted back into the holster, the adapter is automatically unlocked from the connector and the adapter is locked to the holster.

(43) In an embodiment, the EVSE locks the adapter to the holster with a spring-loaded latch controlled by a linearly actuated motor. The EVSE detects whether the adapter is present. The EVSE may include a sensor for detecting whether the adapter is present. As examples, the sensor may be a magnetic field sensor (e.g., a Reed switch, a Hall effect sensor), an optical sensor, an infrared sensor, a mechanical leaf or pushbutton switch, an inductive proximity sensor, or a capacitive proximity sensor. In addition to, or in lieu of, a sensor, the EVSE may detect whether the adapter is present through a proximity/pilot connection pin to detect electrical continuity. The EVSE unlocks the adapter only when the EVSE detects that the adapter is present. After a time duration, the latch returns to the locked position. The adapter can be inserted back into the holster with the latch in the locked position by overcoming the spring force.

(44) In an embodiment, the EVSE controls the locking and unlocking of the adapter to the connector using a locking pin contained within the adapter controlled by an electromagnet that is external to the adapter. The locking pin is spring loaded to the locked position. To move the locking pin to an unlocked state, the electromagnet is activated thereby moving the locking pin to an unlocked position. The unlocked position allows the latch of the connector to be unlatched. To return the locking pin to the locked state, the electromagnet is deactivated causing the spring of the locking pin to return the locking pin to the locked state. This solution may be fully sealed (e.g., no potential for liquid ingress) with no holes for mechanical mechanisms or electrical signal points. The adapter is passively locked with no powered components in the adapter.

(45) In an embodiment, the EVSE controls the locking and unlocking of the adapter to the connector using a locking pin contained within an adapter with an undercut slot for a ball-lock quick release. The locking pin is spring loaded to the locked position. The EVSE has a ball-lock quick release pin with linear actuation. The EVSE pin engages the adapter locking pin and overcomes the spring force to move the locking pin to an unlocked position thereby unlocking the latch of the connector.

(46) In an embodiment, the EVSE controls the locking and unlocking of the adapter to the connector using a latch with an attached magnet that is controlled by a magnet that is external to the adapter. The adapter latch can be spring loaded to the locked position. The EVSE has a linearly actuated motor that controls the movement of the external magnet. To unlock the connector from the adapter, the EVSE moves the external magnet to a position such that the magnetic force of the external magnet attracts the attached magnet of the latch to prevent the latch of the adapter from latching to the latch of the connector. In such an embodiment, actuating the connector latch may be used to terminate a charging session and/or trigger the opening of a door of an electric vehicle inlet.

(47) The determination to use the adapter can be done in different ways. The EVSE may receive a communication originating from the electric vehicle operator that specifies or indicates an intention to use the adapter. For example, the electric vehicle operator may specify the type of vehicle to be charged (e.g., year, model, make) and the EVSE may determine whether that type of vehicle needs the adapter. As another example, the electric vehicle operator may specify the type of connector

their vehicle uses. The electric vehicle operator may provide this information using a mobile application, using a web application, using an on-board infotainment system on their vehicle, using a user interface of the EVSE and/or pressing the latch button of the connector for a predetermined duration and/or a number of times within a predetermined period. The EVSE may receive this communication from a server, from a communication device of the vehicle operator (e.g., a WLAN or WPAN device), from the electric vehicle (e.g., wireless connection between the electric vehicle and the EVSE), or directly from a user interface of the EVSE. This information can be provided at the time of charging or may be provided prior to charging and saved in a profile associated with the vehicle and/or operator. This information may be provided as part of a request for charging service. As another example, the EVSE may perform a vehicle recognition (e.g., using a camera on the EVSE) and determine whether the adapter is needed.

(48) In an embodiment the EVSE may include multiple charging cables and multiple charging connectors. If multiple charging cables and connectors are connected to their respective holsters, the EVSE controls the locking or unlocking of the adapter for each of these multiple connectors if the EVSE does not know which cable the vehicle operator intends to use. For example, if an adapter is to be used, the adapter for each corresponding charging connector is locked to that connector and each of those adapters is unlocked from its respective holster. The unused one or more charging connector with adapters are each put back into its idle state (e.g., those adapter(s) are locked to their respective holster). Determining the unuse or use of a particular charging connector may be done by a timer expiring, a signal received when one of the charging connectors with an adapter is removed from the EVSE, or a signal received when one of the charging connectors with an adapter is inserted into an electric vehicle.

(49) If the adapter is lost or stolen (e.g., removed from the connector after removal of the connector with the adapter attached), the EVSE may record the identifier of the vehicle operator that used the adapter.

(50) FIG. 1 shows a portion of an electric vehicle supply equipment (EVSE) **110** that includes an adapter that is nested within the EVSE **110** according to an embodiment. The EVSE **110** shown FIG. 1 shows two charging cables and connectors for charging (the cable **104** with the connector **115**, and the cable **106** with the connector **135**). However, the number of charging cables and connectors is exemplary as the EVSE may include a single cable and connector or more than two charging cables and connectors. The EVSE **110** includes a component that holds an adapter when not in use and can also hold the attached connector when not in use. This component is sometimes referred herein as a “holster.” There can be a separate holster for each separate charging cable and connector. As illustrated in FIG. 1, the holster **125** and the holster **130** are designed to nest the charging connector adapter **120** and **132** respectively. The charging connector adapter **132** is shown—as being nested within the holster **130**. The charging connector adapter **120** is shown as being attached to the connector **115**.

(51) FIG. 2 is a front view of an exemplary charging connector **115** according to an embodiment. FIG. 3 is a perspective side view of the exemplary charging connector **115** of FIG. 2. The exemplary charging connector **115** is compliant with the Combined Charging System (CCS) Combo 1 standard. The connector **115** includes a connector latch **210**. The connector latch **210** may be used to latch and unlatch the connector **115** to vehicle inlets. As will be described in greater detail later herein, the connector latch **210** is used to lock an adapter **120** on the connector latch **210**. The connector **115** includes a number of contacts that mate with corresponding conductive components (e.g., pins) of a vehicle inlet. Also, the connector **115** is designed to mate with the adapter **120**. As illustrated in FIGS. 2 and 3, the connector **115** includes the proximity detection contact **215**, the ground contact **217**, the pilot contact **219**, the power contact **221**, and the power contact **223**.

(52) The proximity detection contact **215** carries a proximity signal that is used by an electric vehicle to detect the presence of the connector **115** in its electric vehicle charging inlet. Proximity

detection may be tied to a drive interlock system on the electric vehicle to prevent the electric vehicle from being started while the connector **115** is plugged in the electric vehicle. The proximity detection contact **215** is connected to a proximity detection circuit in the connector **115**. The ground contact **217** is a grounding conductor between the EVSE **110** and an electric vehicle (when plugged into the electric vehicle charging inlet of that electric vehicle). The ground contact **217** provides a return path for the control pilot and a reference for the control pilot and proximity voltage measurements.

(53) The pilot contact **219** is used to carry a control pilot signal from the EVSE **110**. The pilot signal may be used to determine whether an electric vehicle is ready to accept energy, to communicate the charging station's maximum available current capacity, determine whether indoor ventilation is required, and verify the grounding circuitry.

(54) FIG. **4** is a perspective view of a first side of the adapter **120** according to an embodiment. FIG. **5** is a front view of the first side of the adapter **120**. The first side of the adapter **120** is the side that mates with the connector **115** and is referred herein as the adapter inlet **405**. The adapter inlet **405** includes conductive components (e.g., pins) that mate with corresponding contacts of the connector **115**. The adapter inlet **405** includes the conductive component **412** that mates with the pilot contact **219**, the conductive component **414** that mates with the ground contact **217**, the conductive component **416** that mates with the proximity detection contact **215**, the conductive component **418** that mates with the power contact **223**, and the conductive component **420** that mates with the power contact **221**. The adapter inlet **405** is compliant with the CCS1 standard. The adapter **120** also includes the adapter locking pin **410**. The EVSE **110** controls the locking and unlocking of the adapter **120** to the connector **115** using the adapter locking pin **410**. In an embodiment, the adapter locking pin **410** is used to lock and unlock the connector latch **210** of the connector **115**.

(55) FIG. **6** is a perspective view of a second side of the adapter **120** according to an embodiment. FIG. **7** is a front view of the second side of the adapter **120**. The second side of the adapter **120** is the side that connects with an electric vehicle inlet when charging an electric vehicle, and connects with an adapter housing of the EVSE. This side of the adapter **120** includes the housing **605** that houses the adapter connector **610**. The adapter connector **610** includes a number of contacts that mate with corresponding conductive components (e.g., pins) of a vehicle inlet. Also, the adapter connector **610** is designed to mate with an adapter housing of the EVSE when not in use. The adapter connector **610** includes the power contact **612**, the power contact **614**, the proximity pilot contact **616**, the ground contact **618**, and the control pilot contact **620**. The power contact **612** and the power contact **614** are used to carry power. The adapter connector **610** also includes a cutout **622** that is used for locking the adapter connector **610** into a vehicle inlet and used for locking the adapter connector **610** into the holster. The adapter connector **610** is compliant with the North American Charging Standard.

(56) The proximity pilot contact **616** is used to determine the status of the adapter connector **610** in an electric vehicle charging inlet (e.g., disconnected, latched, unlatched). Proximity detection may be tied to a drive interlock system on the electric vehicle to prevent the electric vehicle from being started while the adapter connector **610** is plugged in the electric vehicle.

(57) The ground contact **618** is a grounding conductor between the EVSE **110** and an electric vehicle (when plugged into the electric vehicle charging inlet of that electric vehicle). The ground contact **618** provides a return path for the control pilot.

(58) The control pilot contact **620** is used to carry a control pilot signal from the EVSE **110**. The control pilot signal is used as a digital communication path between the EVSE **110** and the electric vehicle. The control pilot signal may also be used to determine whether an electric vehicle is ready to accept energy, to communicate the charging station's maximum available current capacity, determine whether indoor ventilation is required, and verify the grounding circuitry

(59) The adapter **120** electrically connects the conductive components of the first side of the

adapter **120** with the contacts of the second side of the adapter **120**. FIG. **8** is a first view of the internal components of the adapter **120** for electrically connecting the conductive components of the first side of the adapter **120** with the contacts of the second side of the adapter **120**. FIG. **9** is a second view of the same. FIGS. **8** and **9** do not show the enclosure of the first side or the enclosure of the second side of the adapter **120**. The adapter **120** includes the busbar **810** and the busbar **815**. The busbar **810** electrically connects the power conductive component **418** with the power contact **612**. The busbar **815** electrically connects the power conductive component **420** with the power contact **614**. The conductive component **412** is electrically connected to the control pilot contact **620** and carries the control pilot signal. The conductive component **414** is electrically connected to the ground contact **618**. The conductive component **416** is electrically connected to the proximity pilot contact **616** and is used to determine proximity.

(60) FIG. **10** is a perspective view of the adapter **120** connected to the connector **115** according to an embodiment. FIG. **11** is a side view of the adapter **120** connected to the connector **115**. As can be seen in these Figures, the end of the connector **115** fits within the adapter **120** and the adapter connector **610** extends from the adapter **120**. FIG. **11** also shows the button **105** that can be pushed by a user to release the latch **210**. FIG. **10** shows an embodiment where the locking pin of the adapter **120** is designed to interact with an external electromagnet. In an embodiment where the locking and unlocking of the adapter to the connector uses a locking pin that is engaged with an external pin (e.g., FIGS. **20-23**), the adapter **120** includes an opening for the insertion and removal of the external pin.

(61) FIG. **12** shows a holster arrangement that includes two holsters that are each capable of holding a connector when not in use according to an embodiment. Although the holster arrangement **1210** includes two holsters (one for each charging connector), the number of holsters is exemplary as the EVSE may include a single charging cable and connector and a single holster, or more than two charging cables and connectors and more than two corresponding holsters. For illustrative purposes, the adapter **120** is shown as being nested within the holster **125** and there is no adapter shown nested in the holster **130**.

(62) FIG. **13** shows a side view of a holster **125** according to an embodiment and FIG. **14** shows the side view of the holster **125** without the holster housing **1305**. The adapter **120** nests within the holster **125**. The holster **125** includes the adapter locking unit **1310** to lock and unlock the adapter **120** to and from the holster **125**. When the adapter **120** is not being used (e.g., it is not to be connected to the connector **115**), the adapter locking unit **1310** locks the adapter **120** to the holster **125**. This prevents the adapter **120** from being removed from the holster **125** for unauthorized purposes.

(63) The adapter locking unit **1310** includes the latch **1410** that is a spring-loaded latch that is controlled by the linearly actuated motor **1415**. The adapter locking unit **1310** includes the compression spring **1412**, the compression spring **1414**, and the compression spring **1416**. The latch **1410** is in an extended (locked position) to lock the adapter **120** to the holster **125**. The latch **1410** extends through the cutout **622** of the adapter **120** to lock the adapter **120**. In an embodiment, the EVSE **110** only unlocks the adapter **120** (e.g., retract the latch **1410**) when the EVSE **110** detects (e.g., with a sensor and/or through a pilot signal/proximity signal) that the adapter **120** is present. As examples, the sensor may be a magnetic field sensor (e.g., a Reed switch, a Hall effect sensor), an optical sensor, an infrared sensor, a mechanical leaf or pushbutton switch, an inductive proximity sensor, or a capacitive proximity sensor.

(64) The EVSE **110** may return the latch **1410** to a locked position after a time duration. The adapter **120** can be inserted back to the holster **125** with the latch **1410** in a locked position by overcoming the spring force. For example, referring to FIG. **12**, the latch **1232** is in a locked position and there is no adapter present in the holster **130**.

(65) FIG. **15** shows a cross-section view of the side view of FIG. **13** when the latch **1410** is in a locked position and FIG. **16** shows a cross-section view of the side view of FIG. **13** when the latch

1410 is in an unlocked position. As shown in FIG. **15**, the latch **1410** extends within the cutout **622**. This prevents the adapter **120** from being removed from the holster **125**. As shown in FIG. **16**, the latch **1410** does not extend to the cutout **622**. This allows the adapter **120** to be removed from the holster **125**.

(66) The EVSE **110** also locks and unlocks the adapter **120** to and from the connector **115**. In an embodiment, the EVSE **110** controls the locking and unlocking of the adapter to and from the connector using a locking pin contained within the adapter controlled by an electromagnet that is external to the adapter. FIG. **17** illustrates a portion of a cross-section of the connector **115**, adapter **120**, and holster **125** that shows the locking control using a locking pin **410** contained within the adapter **120** controlled by the electromagnet **1710** that is external to the adapter **120**. The electromagnet **1710** is part of the holster **125**. The locking pin **410** is attached to a locking pin plunger **1707**. The locking pin plunger **1707** and the locking pin **410** are contained within the adapter **120**.

(67) The locking pin **410** is spring loaded to the locked position (where the latch **210** of the connector **115** is prevented from being removed from the notch **1705** of the adapter **120**). The connector **115** is locked to the adapter **120** when inserted into the adapter **120**. When inserted, the latch **210** pushes against the locking pin **410** causing the compression spring **1706** to compress until the hook portion **1708** of the latch **210** snaps down to the notch **1705** and clearing the locking pin **410**. The compression spring **1706** then decompresses causing the locking pin **410** to move back to the locked position.

(68) The electromagnet **1710** controls the movement of the locking pin plunger **1707** and thus the locking pin **410**. For example, when activated, the electromagnet **1710** pulls back the locking pin plunger **1707** overcoming the spring force of the spring **1706** that causes the locking pin **410** to be put in an unlocked position where the connector **115** can be removed from the adapter **120**. For example, the user can compress the button **105** to lift the hook portion **1708** of the latch **210** out of the notch **1705** and remove the connector **115** from the adapter **120**.

(69) FIG. **18** shows the locking pin plunger **1707** pulled back by the electromagnet **1710** (the electromagnet **1710** is not shown in FIGS. **18** and **19**) thereby putting the locking pin **410** in an unlocked position where the latch **210** of the connector **115** can be removed from the adapter **120**. FIG. **19** shows the locking pin plunger **1707** in a locking position (not pulled back by the electromagnet **1710**) thereby putting the locking pin **410** in a locked position where the latch **210** of the connector **115** cannot be removed from the adapter **120**. The embodiment shown in FIGS. **17-19** can be fully sealed with no holes for mechanical mechanisms or electrical signal points.

(70) In another embodiment, the EVSE **110** locks and unlocks the adapter **120** to and from the connector **115** using an adapter locking pin contained with the adapter with an undercut slot for a ball-lock quick release. FIGS. **20-23** show such an exemplary embodiment. FIG. **20** shows a cross-section view of the connector **115**, the adapter **120**, and the holster **125** according to an embodiment. The adapter locking pin **2005** is spring loaded to the locked position. The connector **115** is locked to the adapter **120** when inserted into the adapter **120**. When inserted, the latch **210** pushes against the adapter locking pin **2005** causing the compression spring **2006** to compress until the hook portion **1708** of the latch **210** snaps down to the notch **2004** and clearing the adapter locking pin **2005**. The compression spring **2006** then decompresses causing the adapter locking pin **410** to move back to the locked position. The holster **125** includes a ball-lock quick release pin **2007** that is linear actuated. As shown in FIG. **20**, the ball-lock quick release pin **2007** is inserted via a linear actuator (the linear actuator is not shown in FIG. **20**). FIG. **21** shows the cross-section view of FIG. **20** where the ball-lock quick release pin **2007** engages the adapter locking pin **2005** and overcomes the spring force of the compression spring **2006** to move the adapter locking pin **2005** to an unlocked position thereby allowing the connector **115** to be removed from the adapter **120**. FIG. **22** shows the cross-section view of FIG. **20** where once a hard stop is hit, the ball-lock quick release pin **2007** disengages from the adapter locking pin **2005**. After disengaging, the

adapter locking pin **2005** springs back to the locked position. FIG. **23** shows the cross-section view of FIG. **20** where after disengaging, the compression spring **2006** returns the adapter locking pin **2005** to a locked position. The embodiment shown in FIGS. **20-23** requires a specific unlocking pin with ball-locks to defeat the adapter locking mechanism.

(71) In another embodiment, the EVSE **110** locks and unlocks the adapter **120** to and from the connector **115** using a latch with an attached magnet that is controlled by a magnet that is external to the adapter **120**. FIGS. **24-29** show such an exemplary embodiment. In such an embodiment, the connector **115** can be inserted into the adapter **120** even if the adapter **120** is locked to the holster **125**. Further, the connector latch can be actuated even if the connector **115** is locked to the adapter **120**. In an embodiment, the actuation of the connector latch may trigger the termination of the charging session and/or be used to trigger the unlocking and/or opening of the charging port door of an electric vehicle.

(72) FIG. **24** shows a perspective view of the first side of the adapter **120** according to an embodiment where the adapter **120** is locked and unlocked from the connector **115** using a latch with an attached magnet or magnetic coil. In this embodiment, the adapter **120** includes a sloping surface **2410** and a latch **2420**. FIG. **25** shows an exemplary latch **2420** according to an embodiment. The latch **2420** includes the internal magnet **2510**. The internal magnet **2510** can interact with another magnet or magnetic coil that is external to the adapter **120** (e.g., shown in FIG. **27**). The latch **2420** includes a hook end **2515** that can hook on the hook portion of the latch of the connector **115**. The latch **2420** includes the torsion springs **2525** and **2530** that wrap around the pin **2520** to follow the latch of the connector **115** unless the magnetic force from the external magnet or magnetic coil is present.

(73) FIG. **26** shows a cross-section view of the adapter **120** of FIG. **24** according to an embodiment. FIG. **27** shows a cross-section view of the connector **115**, adapter **120**, and the holster **125** that shows the locking control using a latch with an attached magnet. Although FIG. **27** shows an attached magnet, a magnetic coil may also be used. In the embodiment shown in FIG. **27**, the holster **125** includes an external magnet **2710** that slides via a linearly actuated motor **2714**. For example, when the connector **115** is not locked to the adapter **120**, the linear actuated motor **2714** causes the external magnet **2710** to be in a position such that the magnetic force attracts the internal magnet **2510**. In this unlocked position, the latch **2420** is prevented from hooking to the hook portion **1708** of the latch **210** of the connector **115**. This allows the removal of the connector **115** from the adapter **120**. For example, the user can compress the button **105** to lift the hook portion **1708** of the latch **210** and the magnetic force from the external magnet **2710** that is applied to the internal magnet **2510** prevents the latch **2420** from traveling with the latch **210**. FIG. **28** shows the latch **2420** in an unlocked position thereby allowing the connector **115** to be removed from the adapter **120**. In FIG. **28**, the external magnet **2710** (which is not shown in FIGS. **28** and **29**) is in a position that attracts the internal magnet **2510** thereby preventing the latch **2420** from hooking to the hook portion **1708** of the latch **210** of the connector **115**.

(74) When the connector **115** is locked to the adapter **120**, the linear actuated motor **2714** causes the external magnet **2710** to be in a position such that the magnetic force does not attract the internal magnet **2510**. In this locked position, the hook end **2515** of the latch **2420** hooks on the hook portion **1708** of the latch **210** such that the connector **115** is locked to the adapter **120**. For example, the user compresses the button **105** to lift the hook portion **1708** of the latch **210** and since the magnetic force from the external magnet **2710** is not present, latch **2420** travels with the latch **210**. FIG. **29** shows the latch **2420** in a locked position (the magnetic force from the magnet **2710** is not present) thereby locking the connector **115** to the adapter **120**.

(75) The embodiment shown in FIGS. **24-29** can be fully sealed with no holes for mechanical mechanisms or electrical signal points.

(76) Although FIGS. **24-29** describe an attached magnet (the internal magnet **2510**), the attached magnet may be replaced with a ferromagnetic metal. In such a case, the external magnet **2710**

provides a stronger magnetic force to attract and hold the latch **2420**.

(77) The embodiment shown in FIGS. **24-29** allows the latch **210** to operate (move up and down) when the connector **115** is inserted into the adapter **120**. The EVSE **110** may detect when the latch button **105** is pressed and take one or more actions. The one or more actions can be different based on the state of the adapter **120** when the latch button **105** is pressed, the duration of the press of the latch button **105**, and/or the frequency within a period in which the latch button **105** is pressed.

(78) As an example, the EVSE **110** may detect that the latch button **105** is pressed when the connector **115** is attached to the adapter **120**, the pair is removed from the holster **125**, and the adapter **120** is not connected to an electric vehicle. Responsive detecting this condition, the EVSE **110** can take an action including transmitting an RF signal to unlock and/or open a charging port door of an electric vehicle inlet. In such an example, the latch button **105** is being used for an unconventional purpose (the triggering of an RF signal to unlock and/or open a charging port door).

(79) As another example, the EVSE **110** may detect that the latch button **105** is pressed when the connector **115** is attached to the adapter **120** and the adapter **120** is connected to an electric vehicle. Responsive to detecting this condition, the EVSE **110** can take an action including terminating a charging session. In such an example, the latch button **105** is being used for an unconventional purpose (the termination of the charging session).

(80) As another example, the EVSE **110** may detect that the latch button **105** is pressed when the connector **115** is attached to the adapter **120** that is itself in the holster **125**. In such a situation, the EVSE **110** may perform an action for unlocking the adapter **120** from the holster **125**, and/or an action for unlocking the connector **115** from the adapter **120**, which may depend on a duration of the press of the latch button **105** and/or the frequency within a period in which the latch button **105** is pressed. As an example, the EVSE **110** may be configured to unlock the adapter **120** from the holster **125** if the EVSE **110** detects the latch button **105** is pressed down greater than a period (e.g., a 'long' press). As another example, the EVSE **110** may be configured to unlock the connector **115** from the adapter **120** responsive to detecting that the latch button **105** is pressed a number of times in succession during a period (e.g., double pressing).

(81) In the embodiment shown in FIGS. **24-29**, the adapter **120** may include a circuit that is used by the EVSE **110** when detecting states of the adapter **120**. FIG. **30A** shows an exemplary circuit schematic of such a circuit according to an embodiment. The voltage of the proximity signal is measured at the ADC **3007**. The connector **115** includes a proximity detection circuit **3010**. The switch **3012** is mechanically linked to the latch of the connector **115**. The switch **3012** is normally closed except when the latch is actuated. The adapter **120** includes a circuit **3020** that includes a Zener diode **3022**, Schottky diode **3024**, and a resistor **3021** in series between the pilot pin and the proximity pin. Although the exemplary circuit shows a single Zener diode, there may be multiple Zener diodes in series (e.g., two Zener diodes in series). For example, one Zener diode may have a positive temperature coefficient and the other Zener diode may have a negative temperature coefficient that cancel out to provide a stable voltage drop over a wide range of temperatures. The Schottky diode **3024** blocks current from the proximity line to the control pilot line. The ferrite bead **3026** is used to filter all high-frequency current that may pass through the Zener diode **3022** and the Schottky diode **3024**. The ferrite bead **3026** is optional in an embodiment. The circuit **3020** also includes the thermal cutoff (TCO) **3027** on the proximity pilot line and the TCO **3028** on the control pilot line. Although FIG. **30A** shows a TCO on the proximity pilot line and the control pilot line, in an embodiment the thermal cutout could be only on the proximity pilot line or only on the control pilot line. Further, the circuit **3020** can include a resistor in parallel to the TCO **3027** on the proximity pilot line for derating the charger. In such a case, if the TCO **3027** opens, the resistor would change the proximity voltage and the EVSE and the electric vehicle can translate that to an over temperature event. The vehicle inlet **3050** and the vehicle **3055** include circuits for proximity detection (which are not changed in embodiments described herein). Apart of the circuit **3020** is represented by the group **3025**, which includes the resistor **3021**, the Zener diode **3022**, the

Schottky diode **3024**, and the ferrite bead **3026**.

(82) FIG. **30B** shows a circuit that replaces part of the circuit **3020** of FIG. **30A** according to an embodiment. In particular, the group **3025** of FIG. **30A** is replaced with the circuit **3030** shown in FIG. **30B**. The rest of the circuitry shown in FIG. **30A** remains. The circuit shown in FIG. **30B** may protect against a failure of the Zener diode **3022** of FIG. **30A** to prevent safety risks. The circuit **3030** includes the Zener diodes **3031** and **3032** with positive and negative thermal coefficients to compensate for thermal voltage drift. However, the Zener diodes **3031** and **3032** could be replaced with one Zener diode. The Zener diode **3035** and gate to source voltage of the MOSFET **3033** are selected such that the sum of voltage drops on the Zener diode **3035** and the V_{gs} threshold is slightly less than the voltage drops on the Zener diodes **3031** and **3032**. Thus, in normal conditions, the MOSFET **3033** is on and the circuit **3030** acts the same as the circuit shown in FIG. **30A**. However, if the Zener diodes **3031** and/or **3032** fail, the voltage on the gate to source voltage decreases below the threshold voltage and the MOSFET **3033** goes into saturation region. In this case, the voltage drop on the MOSFET drain source is almost equal to the failed (shorted) Zener diode(s) and the system will continue to work as expected. The voltage drop is never less than the sum of voltage drops on the Zener diode **3035** and the V_{gs} threshold of the MOSFET **3033**. The resistor **3036** discharges the gate-source capacitor of the MOSFET **3033** when the failure of the Zener diodes **3031** and/or **3032** occurs. The resistor **3034**, which is optional, limits the gate current.

(83) The EVSE **110** can detect whether the connector **115** is connected to the adapter **120** and the adapter **120** is not connected to an electric vehicle based on the voltage of the control pilot signal (e.g., greater than 10 volts and lower than 11 volts). If the connector **115** is attached to the adapter **120** and the adapter **120** is plugged into an EV, the voltage of the control pilot signal drops (e.g., less than 10 volts) which causes the Zener diode **3022** and the Schottky diode **3024** to turn off. If the connector **115** is attached to the adapter **120** and the adapter **120** is not connected to an EV, the EVSE **110** can detect when the latch of the connector **115** is actuated by detecting a raise in the voltage of the proximity signal (e.g., raise from 0.25 volts to 0.7 volts). In an embodiment, responsive to detecting the latch of the connector **115** is actuated when the connector **115** is attached to the adapter **120**, removed from the holster, and not connected to the EV, the EVSE **110** causes an RF signal to be transmitted to unlock and/or open a charging port door of the vehicle inlet. The RF transmitter may be included in a circuit of the EVSE **110**. The RF transmitter may also be included in a circuit in the connector **115** or the adapter **120**.

(84) The determination of the states includes reading the voltages of the pilot signal and the proximity signal and may include reading a presence sensor (e.g., a magnetic field sensor). FIG. **31** is a table that shows exemplary states according to an embodiment. The state 1A has the connector **115** attached to the adapter **120** and plugged into the holster **125**, and the latch button **105** of the connector **115** not pressed. The EVSE **110** may determine state 1A when the voltage of the pilot signal is greater than 10 volts and lower than 11 volts (e.g., 10.5 volts), the voltage of the proximity signal is between 0.1 volts and 0.5 volts (e.g., 0.25 volts), the presence sensor indicates the adapter is 120 is within the holster inlet of the holster **125**. In state 1A, the latch button **105** of the connector **115** is not pressed.

(85) State 1B is where the connector **115** is attached to the adapter **120** and plugged into the holster **125**, and the latch button **105** of the connector **115** is pressed. The EVSE **110** may determine state 1B when the voltage of the pilot signal is greater than 10 volts but under 11 volts (e.g., 10.5 volts), the voltage of the proximity signal is between 0.5 volts and 1 volt (e.g., 0.7 volts), and the presence sensor indicates that the adapter **120** is within the holster inlet of the holster **125**.

(86) The EVSE **110** may, responsive to determining state 1B, take one or more actions including unlocking the adapter **120** from the holster **125** and/or unlocking the connector **115** from the adapter **120** depending on the duration of the press of the latch button **105** and/or the frequency of presses of the latch button **105** within a period. The EVSE **110** may determine the duration of the press of the latch button **105** by measuring the voltage of the proximity signal over a predefined

period. For example, a long press may be determined if the voltage of the proximity signal is between 0.5V to 1V over a predetermined duration. As another example, a short press may be determined if the voltage of the proximity signal raises to between 0.5V to 1V for less than the predetermined duration (e.g., it goes back down to 0.1V to 0.5V). As another example, a double press may be determined if the voltage of the proximity signal raises to between 0.5V to 1V for less than a first predetermined duration (e.g., it goes back down to 0.1V to 0.5V) and the voltage of the proximity signal again raises to between 0.5V to 1V within a predetermined period of the prior detected press. As an example, the EVSE **110** may be configured to unlock the adapter **120** from the holster **125** if the EVSE **110** detects the latch button **105** is pressed down greater than a duration (e.g., a 'long' press). As another example, the EVSE **110** may be configured to unlock the connector **115** from the adapter **120** responsive to detecting that the latch button **105** is pressed a number of times in succession during a period (e.g., double pressing).

(87) State 2A is where the connector **115** is attached to the adapter **120** and the adapter **120** is not within the holster **125**, and the latch button **105** of the connector **115** is not pressed. The EVSE **110** may determine state 2A when the voltage of the pilot signal is greater than 10 volts but under 11 volts (e.g., 10.5 volts), the voltage of the proximity signal is between 0.1 volts and 0.5 volts (e.g., 0.25 volts), and the presence sensor indicates that the adapter **120** is not within the holster inlet of the holster **125**.

(88) State 2B is where the connector **115** is attached to the adapter **120** and the adapter **120** is not within the holster **125**, and the latch button **105** of the connector **115** is pressed. The EVSE **110** may determine state 2B when the voltage of the pilot signal is greater than 10 volts but under 11 volts (e.g., 10.5 volts), the voltage of the proximity signal is between 0.5 volts and 1 volt (e.g., 0.7 volts), and the presence sensor indicates that the adapter **120** is not within the holster inlet of the holster **125**. The EVSE **110** may, responsive to determining state 2B, transmit an RF signal to trigger the unlocking and/or opening of a charging port door of an electric vehicle inlet.

(89) State 3 is where the connector **115** is not attached to the adapter **120** and the adapter **120** is plugged into the holster **125**. The EVSE **110** may determine state 3 when the voltage of the pilot signal is between 11 volts and 13 volts (e.g., 12 volts), the voltage of the proximity signal is less than 0.1 volts, and the presence sensor indicates the adapter **120** is within the holster inlet of the holster **125**.

(90) State 4 is where the connector **115** is not attached to the adapter **120** and the adapter **120** is not plugged into the holster **125** (e.g., the adapter **120** may be lost). The EVSE **110** may determine state 4 when the voltage of the pilot signal is between 11 volts and 13 volts (e.g., 12 volts), the voltage of the proximity signal is less than 0.1 volts, and the presence sensor includes that the adapter **120** is not within the holster inlet of the holster **125**.

(91) State 5A is where the connector **115** is attached to the adapter **120** and it is plugged into an electric vehicle. The EVSE **110** may determine state 5A when the voltage of the pilot signal is less than 10 volts and the voltage of the proximity signal is between 1 volt to 2 volts.

(92) State 5B is where the connector **115** is attached to the adapter **120** and it is plugged into an electric vehicle, and the latch button **105** of the connector **115** is pressed. The EVSE **110** may determine state 5B when the voltage of the pilot signal is less than 10 volts and the voltage of the proximity signal is above 2 volts. The EVSE **110** may, responsive to determining state 5B, cause a charging session to be terminated.

(93) FIG. **32** is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment. The operations of FIG. **32** are described with respect to the exemplary embodiments shown in FIGS. **1-29**. However, the operations of FIG. **32** can be performed by embodiments different from those shown in FIGS. **1-29**, and the embodiments shown in FIGS. **1-29** can perform operations different from those described in FIG. **32**.

(94) The EVSE **110** includes at least one charging cable **104** that terminates with a charging cable connector **115** and an adapter **120** that is nested within a holster **125**. The adapter **120** allows a

different type of charging connector to be used. As an example, the charging cable connector **115** may be CCS1 compliant and the adapter **120** may be a CCS1-to-NACS adapter. The holster **125** may include an inlet to receive the CCS1-to-NACS adapter.

(95) At operation **3210**, the EVSE **110** locks a charging connector adapter **120** to a holster **125** of the EVSE **110**. The charging connector adapter **120** is nested within the holster of the EVSE. The holster **125** may include an adapter locking unit **1310** to lock and unlock the charging connector adapter **120** to and from the holster **125**. The adapter locking unit **1310** may include a spring-loaded latch that is controlled by a linearly actuated motor **1415**. As a condition for unlocking the adapter **120** (e.g., retracting the latch **1410**), the EVSE **110** must detect (e.g., with a sensor and/or through a pilot signal/proximity signal) that the charging connector adapter **120** is present. As examples, the sensor may be a magnetic field sensor (e.g., a Reed switch, a Hall effect sensor), an optical sensor, an infrared sensor, a mechanical leaf or pushbutton switch, an inductive proximity sensor, or a capacitive proximity sensor. The charging connector adapter **120** can be inserted back into the holster **125** with the latch **1410** in a locked position by overcoming the spring force. Thus, the EVSE **110** may lock the charging connector adapter **120** responsive to the charging connector adapter **120** being inserted into the holster **125**.

(96) Next, at operation **3215**, the EVSE **110** determines whether an electric vehicle that is to be charged at the EVSE **110** through the charging cable **104** with the charging cable connector **115** is to use the adapter **120**. The determination whether to use the adapter **120** can be done in different ways. For instance, the electric vehicle operator may specify or indicate their need to use the adapter **120**. For example, the electric vehicle operator may specify the type of vehicle to be charged (e.g., year, model, make) and the EVSE **110** may determine based on the type of vehicle whether the vehicle needs the adapter **120** for charging. As another example, the electric vehicle operator may specify the type of connector their vehicle uses (e.g., whether it corresponds to the adapter **120** or the connector **115**). The electric vehicle operator may provide this information using a mobile application, using a web application, using an on-board infotainment system on their vehicle, using a user interface of the EVSE, and/or pressing the latch button **105** for a predetermined duration and/or a number of times within a predetermined period. This information can be provided at the time of charging or may be provided prior to charging and saved in a profile associated with the vehicle and/or operator. As another example, the EVSE may perform a vehicle recognition (e.g., using a camera on the EVSE) and determine whether the adapter **120** is needed.

(97) In the embodiment of FIG. **32**, the connector **115** is locked to the adapter **120** that is itself locked to the holster **125**. As a result, the connector **115** cannot be removed from the holster **125** without the connector **115** unless the connector **115** is unlocked from the adapter **120**.

(98) If the adapter **120** is to be used, then operation **3220** is performed. If the adapter **120** is not to be used, then operation **3225** is performed. At operation **3220** (the adapter **120** is to be used), the EVSE **110** unlocks the adapter **120** from the holster **125**. The EVSE **110** may unlock the adapter **120** from the holster **125** by causing the linearly actuated motor **1415** to retract the latch **1410** thereby releasing the lock. Prior to unlocking the adapter **120**, the EVSE **110** first detects (e.g., with a sensor and/or through a pilot signal/proximity signal) that the adapter **120** is present. As examples, the sensor may be a magnetic field sensor (e.g., a Reed switch, a Hall effect sensor), an optical sensor, an infrared sensor, a mechanical leaf or pushbutton switch, an inductive proximity sensor, or a capacitive proximity sensor. If the connector **115** and/or the adapter **120** is not present, then the EVSE **110** does not unlock the adapter **120** from the holster **125**. Because in this embodiment the connector **115** is already locked to the adapter **120**, the vehicle operator may then remove the connector **115** with the attached adapter **120** and insert the adapter **120** into their vehicle inlet. The adapter **120** is locked again to the holster **125** when the adapter **120** is returned to the holster **125**.

(99) At operation **3225** (the adapter **120** is not to be used), the EVSE **110** unlocks the connector **115** from the adapter **120**. With respect to the embodiments that use an electromagnet to control

movement of a locking pin (e.g., FIGS. 17-19), the EVSE 110 activates the electromagnet to move a locking pin to an unlocked position to allow the connector 115 to be removed from the adapter 120. For example, with respect to FIGS. 17-19, the EVSE 110 activates the electromagnet 1710 to pull back the locking pin plunger 1707 and thus the locking pin 410. This overcomes the spring force of the spring 1706 that causes the locking pin 410 to be in an unlocked position where the connector 115 can be removed from the adapter 120. The adapter 120 may remain locked to the holster 125. With respect to the embodiments that use an adapter locking pin with an undercut slot for a ball-lock quick release (e.g., FIGS. 20-23), the EVSE 110 inserts the ball-lock quick release pin 2007 via a linear actuator to engage the adapter locking pin 2005 and overcome the spring force to move the adapter locking pin 2005 to an unlocked position. This allows the connector 115 to be removed from the adapter 120. The adapter 120 may remain locked to the holster 125. The connector 115 is locked to the adapter 120 again when the connector 115 is inserted back into the holster 125. With respect to the embodiments that use a latch with an attached magnet that is controlled by a magnet that is external to the adapter 120 (e.g., FIGS. 24-29), the EVSE 110 moves the external magnet 2710 to an unlocked position to allow the connector 115 to be removed from the adapter 120. For example, the external magnet 2710 is moved to a position such that the magnetic force of the magnet 2710 attracts the internal magnet 2510, which prevents the latch 2420 from hooking to the hook portion 1708 of the latch 210 of the connector 115. The adapter 120 may remain locked to the holster 125. The connector 115 is locked to the adapter 120 again when the connector 115 is inserted back into the holster 125.

(100) FIG. 33 is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment. The operations of FIG. 33 are described with respect to the exemplary embodiments shown in FIGS. 1-29. However, the operations of FIG. 33 can be performed by embodiments different from those shown in FIGS. 1-29, and the embodiments shown in FIGS. 1-29 can perform operations different from those described in FIG. 33.

(101) In the embodiment of FIG. 33, the connector 115 is not locked to the adapter 120 when the adapter 120 is locked and nested within the holster 125. The connector 115 can be freely removed from the adapter 120 while the adapter 120 is locked to the holster 125. With respect to the embodiments that use an electromagnet to move the locking pin (e.g., FIGS. 17-19), the EVSE 110 keeps current running through the electromagnet 1710 to keep the locking pin plunger 1707 pulled back and the locking pin 410 in an unlocked position. With respect to the embodiments that use an adapter locking pin with an undercut slot for a ball-lock quick release (e.g., FIGS. 20-23), the EVSE 110 keeps the ball-lock quick release pin 2007 engaged with the adapter locking pin 2005 to keep the adapter locking pin 2005 in an unlocked position. With respect to the embodiments that use a latch with an attached magnet that is controlled by a magnet that is external to the adapter 120 (e.g., FIGS. 24-29), the EVSE 110 keeps the external magnet 2710 in a position to apply a magnetic force that attracts the internal magnet 2510 that prevents the latch 2420 from hooking to the latch 210 of the connector 115.

(102) The EVSE 110 includes at least one charging cable 104 that terminates with a charging cable connector 115 and an adapter 120 that is nested within a holster 125. The adapter 120 allows a different type of charging connector to be used. As an example, the charging cable connector 115 may be CCS1 compliant and the adapter 120 may be a CCS1-to-NACS adapter. The holster 125 may include an inlet to receive the CCS1-to-NACS adapter.

(103) At operation 3310, the EVSE 110 locks a charging connector adapter 120 to a holster 125 of the EVSE 110. The operation 3310 is like the operation 3210. The charging connector adapter 120 is nested within the holster of the EVSE. The holster 125 may include an adapter locking unit 1310 to lock and unlock the charging connector adapter 120 to and from the holster 125. The adapter locking unit 1310 may include a spring-loaded latch that is controlled by a linear actuated motor 1415. As a condition for unlocking the adapter 120 (e.g., retracting the latch 1410), the EVSE 110 must detect (e.g., with a sensor and/or through a pilot signal/proximity signal) that the charging

connector adapter **120** is present and the connector **115** is connected to the adapter **120**. As examples, the sensor may be a magnetic field sensor (e.g., a Reed switch, a Hall effect sensor), an optical sensor, an infrared sensor, a mechanical leaf or pushbutton switch, an inductive proximity sensor, or a capacitive proximity sensor. The charging connector adapter **120** can be inserted back into the holster **125** with the latch **1410** in a locked position by overcoming the spring force. Thus, the EVSE **110** may lock the charging connector adapter **120** responsive to the charging connector adapter **120** being inserted into the holster **125**.

(104) Next, at operation **3315**, the EVSE **110** determines whether an electric vehicle that is to be charged at the EVSE **110** through the charging cable **104** with the charging cable connector **115** is to use the adapter **120**. The operation **3315** is like the operation **3215**. The determination whether to use the adapter **120** can be done in different ways. For instance, the electric vehicle operator may specify or indicate their need to use the adapter **120**. For example, the electric vehicle operator may specify the type of vehicle to be charged (e.g., year, model, make) and the EVSE **110** may determine based on the type of vehicle whether the vehicle needs the adapter **120** for charging. As another example, the electric vehicle operator may specify the type of connector their vehicle uses (e.g., whether it corresponds to the adapter **120** or the connector **115**). The electric vehicle operator may provide this information using a mobile application, using a web application, using an on-board infotainment system on their vehicle, using a user interface of the EVSE and/or pressing the latch button **105** for a predetermined duration and/or a number of times within a predetermined period. This information can be provided at the time of charging or may be provided prior to charging and saved in a profile associated with the vehicle and/or operator. As another example, the EVSE may perform a vehicle recognition (e.g., using a camera on the EVSE) and determine whether the adapter **120** is needed.

(105) If the adapter **120** is to be used, then operation **3320** is performed. At operation **3320**, since the connector **115** is not locked to the adapter **120**, the EVSE **110** locks the connector **115** to the adapter **120**. With respect to the embodiments that use an electromagnet to move the locking pin (e.g., FIGS. **17-19**), the EVSE **110** deactivates the electromagnet **1710** to release the locking pin plunger **1707** and cause the spring **1706** to return the locking pin **410** to the locked position. With respect to the embodiments that use a latch with an attached magnet that is controlled by a magnet that is external to the adapter **120** (e.g., FIGS. **24-29**), the EVSE **110** moves the external magnet **2710** to a position such that its magnetic force does not attract the internal magnet **2510**. In such a locked position, the hook end **2515** of the latch **2420** hooks on the hook portion **1708** of the latch **210** such that the connector **115** is locked to the adapter **120**.

(106) Next, at operation **3330**, the EVSE unlocks the adapter **120** from the holster **125**. The EVSE **110** may unlock the adapter **120** from the holster **125** by causing the linearly actuated motor **1415** to retract the latch **1410** thereby releasing the lock. Prior to unlocking the adapter **120**, the EVSE **110** first detects (e.g., with a sensor and/or through a pilot signal/proximity signal) that the adapter **120** is present. As examples, the sensor may be a magnetic field sensor (e.g., a Reed switch, a Hall effect sensor), an optical sensor, an infrared sensor, a mechanical leaf or pushbutton switch, an inductive proximity sensor, or a capacitive proximity sensor. If the adapter **120** is not present, then the EVSE **110** does not unlock the adapter **120** from the holster **125**. The vehicle operator may then remove the connector **115** with the attached adapter **120** and insert the adapter **120** into their vehicle inlet.

(107) When the adapter **120** is returned to the holster **125**, the EVSE **110** unlocks the connector **115** from the adapter **120**. With respect to the embodiments that use an electromagnet to move the locking pin (e.g., FIGS. **17-19**), the EVSE **110** keeps current running through the electromagnet **1710** to keep the locking pin plunger **1707** pulled back and the locking pin **410** in an unlocked position. With respect to the embodiments that use an adapter locking pin with an undercut slot for a ball-lock quick release (e.g., FIGS. **20-23**), the EVSE **110** keeps the ball-lock quick release pin **2007** engaged with the adapter locking pin **2005** to keep the adapter locking pin **2005** in an

unlocked position. With respect to the embodiments that use a latch with an attached magnet that is controlled by a magnet that is external to the adapter **120** (e.g., FIGS. **24-29**), the EVSE **110** moves the external magnet **2710** to an unlocked position to allow the connector **115** to be removed from the adapter **120**. For example, the external magnet **2710** is moved to a position such that the magnetic force of the magnet **2710** attracts the internal magnet **2510**, which prevents the latch **2420** from hooking to the hook portion **1708** of the latch **210** of the connector **115**.

(108) If the adapter **120** is not to be used, since the connector **115** is not locked to the adapter **120**, the connector **115** can be freely removed from the adapter **120**. Therefore, at operation **3325**, no further operations are needed to allow the correct connector to be used for charging the electric vehicle of the operator.

(109) FIG. **34** is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment. The operations of FIG. **34** are described with respect to the exemplary embodiment shown in FIGS. **24-31**.

(110) At operation **3410**, the EVSE **110** detects that the button **105** on the connector **115** has been actuated when the connector **115** is attached to the adapter **120**, the pair is removed from the holster **125**, and the adapter **120** is not connected to an electric vehicle. For example, the EVSE **110** detects a voltage of the control pilot signal in a certain range (e.g., greater than 10 volts and lower than 11 volts (e.g., 10.5 volts)), a voltage of the proximity signal in a certain range (e.g., between 0.5 volts and 1 volts), and a presence sensor (e.g., a magnetic field sensor) indicates that the adapter **120** is not within the holster **125**. Next, at operation **3415**, the EVSE **110** transmits an RF signal that can trigger the opening of a charging port door of an electric vehicle inlet. The RF signal may also trigger the unlocking of the charging port door. The RF transmitter may be included in the EVSE **110**, the connector **115**, or the adapter **120**.

(111) FIG. **35** is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment. The operations of FIG. **35** are described with respect to the exemplary embodiment shown in FIGS. **24-31**.

(112) At operation **3510**, the EVSE **110** detects that the button **105** on the connector **115** has been actuated when the connector **115** is attached to the adapter **120** and the adapter **120** is connected to an electric vehicle. For example, the EVSE **110** detects a voltage of the control pilot signal in a certain range (e.g., less than 10 volts), a voltage of the proximity signal in a certain range (e.g., above 2 volts).

(113) Next, at operation **3515**, the EVSE **110** terminates a charging session that is ongoing. Terminating the charging session may include opening a set of contactors on the EVSE **110** to stop the flow of current.

(114) FIG. **36** is a flow diagram illustrating exemplary operations performed by an EVSE according to an embodiment. The operations of FIG. **36** are described with respect to the exemplary embodiment shown in FIGS. **24-31**.

(115) At operation **3610**, the EVSE **110** detects that the button **105** on the connector **115** has been actuated when the connector **115** is attached to the adapter **120** and the pair is in the holster **125**. As an example, the EVSE **110** detects that the voltage of the pilot signal is greater than 10 volts but under 11 volts (e.g., 10.5 volts), the voltage of the proximity signal is between 0.5 volts and 1 volt (e.g., 0.7 volts), and the presence sensor indicates that the adapter **120** is within the holster inlet of the holster **125**.

(116) Next, at operation **3615**, the EVSE **110** takes one or more actions including unlocking the adapter **120** from the holster **125** and/or unlocking the connector **115** from the adapter **120** depending on the duration of the press of the latch button **105** and/or the frequency of presses of the latch button **105** within a period. The EVSE **110** may determine the duration of the press of the latch button **105** by measuring the voltage of the proximity signal over a predefined period. For example, a long press may be determined if the voltage of the proximity signal is between 0.5V to 1V over a predetermined duration. As another example, a short press may be determined if the

voltage of the proximity signal raises to between 0.5V to 1V for less than the predetermined duration (e.g., it goes back down to 0.1V to 0.5V). As another example, a double press may be determined if the voltage of the proximity signal raises to between 0.5V to 1V for less than a first predetermined duration (e.g., it goes back down to 0.1V to 0.5V) and the voltage of the proximity signal again raises to between 0.5V to 1V within a predetermined period of the prior detected press. As an example, the EVSE **110** may be configured to unlock the adapter **120** from the holster **125** if the EVSE **110** detects the latch button **105** is pressed down greater than a duration (e.g., a 'long' press). As another example, the EVSE **110** may be configured to unlock the connector **115** from the adapter **120** responsive to detecting that the latch button **105** is pressed a number of times in succession during a period (e.g., double pressing).

(117) FIG. **37** shows an exemplary embodiment of an EVSE according to an embodiment. The EVSE **110** may take the form of the EVSE **3700**. Although several components are illustrated as being included in the EVSE **3700**, in some embodiments additional, different, or less components may be used in the EVSE **3700**. For example, some EVSE may not include a display or a user interface. Other EVSE may not include an RFID reader or an energy meter. Other EVSE may include one or more lights that can provide visual indications.

(118) As illustrated in FIG. **37**, the EVSE **3700** includes the holster **3702** (which may include the adapter **3704**), the charging circuitry **3705**, the current control device **3715**, the energy meter **3720**, the volatile memory **3725**, the non-volatile memory **3730** (e.g., hard drive, flash, PCM, etc.), one or more transceiver(s) **3735** (e.g., wired transceiver(s) (e.g., Ethernet, power line communication (PLC), etc.) and/or wireless transceiver(s) (e.g., 802.15.4 (e.g., ZigBee, etc.), RF, Bluetooth, Wi-Fi, Infrared, GPRS/GSM, CDMA, etc.)), the RFID reader **3740**, the display unit **3745**, the user interface **3750**, and the processing system **3755** (e.g., one or more microprocessors and/or a system on an integrated circuit), which may be coupled with one or more buses **3760**.

(119) The holster **3702** holds the adapter **3704** when not in use and can also hold a connector (e.g., a CCS connector) when not in use. The charging circuitry **3705** is circuitry for an attached charging cord (e.g., with a CCS connector). While FIG. **37** illustrates a single holster **3702**, the EVSE **3700** may include multiple charging holsters and which may be the same or different types.

(120) The current control device **3715** controls the current flowing on the power line **3701**. For example, in some embodiments the current control device **3715** energizes the charging circuitry **3705** (e.g., by completing the circuit to the power line **3701**) or de-energizes the charging circuitry **3705** (e.g., by breaking the circuit to the power line **3701**). The current control device **3715** may be a set of contactors. In some embodiments the current control device **3715** energizes the charging circuitry **3705** responsive to receiving a command from a server that indicates charging is authorized.

(121) The energy meter **3720** measures the amount of electricity that is flowing on the power line **3701** through the charging circuitry **3705**. While in one embodiment the energy meter **3720** measures current flow, in an alternative embodiment the energy meter **3720** measures power draw. The energy meter **3720** may be an induction coil or other devices suitable for measuring electricity. While the energy meter **3720** is illustrated as being included within the EVSE **3700**, in other embodiments the energy meter **3720** is exterior to the EVSE **3700** but capable of measuring the amount of electricity flowing on the power line **3701** through the charging circuitry **3705**.

(122) The RFID reader **3740** reads RFID tags from RFID enabled devices (e.g., smartcards, key fobs, contactless credit cards, etc.), embedded with RFID tag(s) of operators that want to use the EVSE **3700**. For example, in some embodiments a vehicle operator can wave/swipe an RFID enabled device near the RFID reader **3740** to provide an access credential for use of the EVSE **3700**.

(123) The transceiver(s) **3735** transmit and receive messages. For example, the transceiver(s) **3735** may transmit authorization requests to the EV charging network server, receive commands from the EV charging network server indicating whether the charging session is authorized, etc. The

transceiver(s) **3735** may include an RF transmitter that can trigger the opening of a charging port door of an electric vehicle inlet as described herein.

(124) The display unit **3745** is used to display messages to vehicle operators including charging status, confirmation messages, error messages, notification messages, etc. The user interface **3750** allows operators to interact with the EVSE **3700**. By way of example, the user interface **3750** allows electric vehicle operators to present an access credential, enter in account and/or payment information, etc. As another example, the user interface **3750** may allow an electric vehicle operator to specify or indicate an intention to use the adapter **3704**.

(125) The processing system **3755** may retrieve instruction(s) from the volatile memory **3725** and/or the non-volatile memory **3730** and execute the instructions to perform operations for the electric vehicle charging station.

(126) Some portions of the present invention may be implemented using code and data stored and executed on the EVSE. The code and/or data may be stored in machine-readable media, such as machine-readable storage media (e.g., magnetic disks; optical disks; random access memory; read only memory; flash memory devices; phase-change memory) and/or machine-readable communication media (e.g., electrical, optical, acoustical or other form of propagated signals-such as carrier waves, infrared signals, digital signals, etc.). Thus, the storage device(s) of a given EVSE typically stores code and/or data for execution on a set of one or more processors of that EVSE. Of course, one or more parts of an embodiment of the invention may be implemented using different combinations of software, firmware, and/or hardware.

(127) In the preceding description, numerous specific details are set forth to provide a more thorough understanding of the present invention. It will be appreciated, however, by one skilled in the art that the invention may be practiced without such specific details. In other instances, control structures, gate level circuits and full software instruction sequences have not been shown in detail in order not to obscure the invention. Those of ordinary skill in the art, with the included descriptions, will be able to implement appropriate functionality without undue experimentation.

(128) References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

(129) While the flow diagrams in the figures show a particular order of operations performed by certain embodiments of the invention, such order is exemplary (e.g., alternative embodiments may perform the operations in a different order, combine certain operations, overlap certain operations, etc.).

(130) While the invention has been described in terms of several embodiments, those skilled in the art will recognize that the invention is not limited to the embodiments described, can be practiced with modification and alteration within the spirit and scope of the appended claims. The description is thus to be regarded as illustrative instead of limiting.

Claims

1. A method in an electric vehicle supply equipment (EVSE), comprising: locking a charging connector adapter to a holster of the EVSE, wherein the charging connector adapter is nested within the holster of the EVSE; determining that an electric vehicle that is to be charged at the EVSE through a charging cable with a charging connector is to use the charging connector adapter; automatically locking the charging connector to the charging connector adapter; automatically

unlocking the charging connector adapter from the holster of the EVSE thereby allowing the charging cable with the charging connector adapter to be removed from the holster and used for connecting the charging connector adapter to an inlet of the electric vehicle; and responsive to the charging connector adapter being inserted back into the holster, locking the charging connector adapter to the holster of the EVSE, and unlocking the charging connector from the charging connector adapter.

2. The method of claim 1, wherein the charging connector is Combined Charging System (CCS) Combo 1 compliant, and wherein the charging connector adapter is adapted to connect to an electric vehicle inlet that is North American Charging Standard (NACS) compliant.

3. The method of claim 1, wherein automatically locking the charging connector to the charging connector adapter includes deactivating an electromagnet that moves a locking pin of the charging connector adapter to a locked position thereby preventing a latch of the charging connector to be unlatched, wherein the electromagnet is external to the charging connector adapter.

4. The method of claim 1, wherein automatically locking the charging connector to the charging connector adapter includes using a linear actuator to disengage a locking pin of the charging connector adapter with a ball-lock quick release pin to move the locking pin to a locked position thereby preventing a latch of the charging connector to be unlatched.

5. The method of claim 1, wherein automatically locking the charging connector to the charging connector adapter includes using a linear actuator to move a first magnet of the holster to a position where a magnetic force of the first magnet does not attract a second magnet that is attached to a latch of the charging connector adapter thereby preventing a latch of the charging connector to be unlatched.

6. The method of claim 1, wherein automatically unlocking the charging connector adapter from the holster of the EVSE includes retracting a spring-loaded latch that latches to the charging connector adapter.

7. The method of claim 1, wherein determining that the electric vehicle that is to be charged at the EVSE through the charging cable with the charging connector is to use the charging connector adapter includes receiving a communication that specifies or indicates an intention to use the charging connector adapter.

8. An electric vehicle supply equipment (EVSE), comprising: a charging connector adapter that is nested within a holster, wherein the charging connector adapter is locked to a charging connector of a charging cable when the charging connector adapter is to be used to couple to an electric vehicle, and wherein the charging connector adapter is unlocked from the charging connector when the charging connector adapter is not to be used to couple to the electric vehicle; the holster including an adapter locking unit to lock and unlock the charging connector adapter to and from the holster; a non-transitory machine-readable storage medium that provides instructions to be executed by a set of one or more processors to perform operations including: responsive to a determination that the charging connector adapter is to be used for charging the electric vehicle, cause the charging connector adapter to be locked to the charging connector, and cause the adapter locking unit to unlock the charging connector adapter from the holster thereby allowing the charging cable with the charging connector adapter to be removed from the holster and used for connecting the charging connector adapter to an inlet of the electric vehicle, responsive to the charging connector adapter being inserted back into the holster, cause the adapter locking unit to lock the charging connector adapter to the holster, and cause the charging connector adapter to unlock from the charging connector.

9. The EVSE of claim 8, wherein the charging connector is Combined Charging System (CCS) Combo 1 compliant, and wherein the charging connector adapter is adapted to connect to an electric vehicle inlet that is North American Charging Standard (NACS) compliant.

10. The EVSE of claim 8, wherein the holster further includes an electromagnet, and wherein the operation to cause the charging connector adapter to be unlocked from the charging connector

includes activating the electromagnet that moves a locking pin of the charging connector adapter to an unlocked position thereby allowing a latch of the charging connector to be unlatched.

11. The EVSE of claim 8, wherein the holster further includes a ball-lock quick release pin that is controlled by a linear actuator, and wherein the operation to cause the charging connector adapter to be unlocked from the charging connector includes causing the linear actuator to insert the ball-lock quick release pin to engage a locking pin of the charging connector adapter and move the locking pin to an unlocked position thereby allowing a latch of the charging connector to be unlatched.

12. The EVSE of claim 8, wherein the holster further includes a first magnet that is controlled by a linear actuator, and wherein the operation to cause the charging connector adapter to be unlocked from the charging connector includes causing the linear actuator to move the first magnet to a position where a magnetic force attracts a second magnet that is attached to a latch of the charging connector adapter and prevents the latch from latching to the charging connector.

13. The EVSE of claim 8, wherein the operation to cause the adapter locking unit to unlock the charging connector adapter from the holster includes causing the adapter locking unit to retract a spring-loaded latch of the adapter locking unit that latches to the charging connector adapter.

14. The EVSE of claim 8, wherein the operation to determine whether the electric vehicle is to be charged using the charging connector adapter includes receiving a communication that specifies or indicates an intention to use the charging connector adapter.

15. A non-transitory machine-readable storage medium that provides instructions that, if executed by a processing system of an electric vehicle supply equipment (EVSE), will cause said EVSE to perform operations comprising: causing a charging connector adapter to be locked to a holster of the EVSE, wherein the charging connector adapter is nested within the holster of the EVSE; responsive to determining that an electric vehicle that is to be charged at the EVSE through a charging cable with a charging connector is to use the charging connector adapter, causing the charging connector to be locked to the charging connector adapter, and causing the charging connector adapter to be unlocked from the holster of the EVSE thereby allowing the charging cable with the charging connector adapter to be removed from the holster and used for connecting the charging connector adapter to an inlet of the electric vehicle; and responsive to the charging connector adapter being inserted back into the holster, causing the charging connector adapter to be locked to the holster of the EVSE, and causing the charging connector to be unlocked from the charging connector adapter.

16. The non-transitory machine-readable storage medium of claim 15, wherein the charging connector is Combined Charging System (CCS) Combo 1 compliant, and wherein the charging connector adapter is adapted to connect to an electric vehicle inlet that is North American Charging Standard (NACS) compliant.

17. The non-transitory machine-readable storage medium of claim 15, wherein automatically locking the charging connector to the charging connector adapter includes deactivating an electromagnet that moves a locking pin of the charging connector adapter to a locked position thereby preventing a latch of the charging connector to be unlatched, wherein the electromagnet is external to the charging connector adapter.

18. The non-transitory machine-readable storage medium of claim 15, wherein automatically locking the charging connector to the charging connector adapter includes using a linear actuator to disengage a locking pin of the charging connector adapter with a ball-lock quick release pin to move the locking pin to a locked position thereby preventing a latch of the charging connector to be unlatched.

19. The non-transitory machine-readable storage medium of claim 15, wherein automatically locking the charging connector to the charging connector adapter includes using a linear actuator to move a first magnet of the holster to a position where a magnetic force of the first magnet does not attract a second magnet that is attached to a latch of the charging connector adapter thereby

preventing a latch of the charging connector to be unlatched.

20. The non-transitory machine-readable storage medium of claim 15, wherein automatically unlocking the charging connector adapter from the holster of the EVSE includes retracting a spring-loaded latch that latches to the charging connector adapter.
